Macroeconomic Drivers of U.S. Unemployment

Forecasting and Explaining U.S. Unemployment Through Macroeconomic Indicators

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1 Title, Team, and Affiliations

Project title: *Forecasting and Explaining U.S. Unemployment Through Macroeconomic Indicators*

Name	Role	Willamette email
Manish R Kallu	Owner Product Lead	mkallu@willamette.edu

Peer Stakeholder POs: Brandon Smith, Jon Garrow, Jackson Garro

Repository: `https://github.com/manishkallu01-wq/DATA510-Session-2`

2 Abstract

(150 to 250 words after you have results. Summarize: problem, approach, key finding, impact, limitation. Write this section last.)

3 Introduction: Problem and Research Questions

Problem statement:

Primary research question:

Secondary questions (if any):

Changes since M1 proposal: *(none yet, or explain pivot)*

4 Impact and Stakeholders

5 Related Work and Background

Wildfire smoke and respiratory health are well studied at regional scale (Reid et al. 2016; Liu et al. 2016). This project applies county-level public data to Oregon specifically.

6 Data and Data Engineering

Sources:

Source	Role	Access

Engineering summary: Raw data in `data/raw/`; processed panel in `data/processed/`; ingest scripts in `src/ingest/`.

```r
# Replace with your project data when ready:
# panel <- readRDS(here::here("data/processed/analysis_panel.rds"))
# nrow(panel)
knitr::kable(data.frame(
  check = c("Rows", "Counties", "Date range"),
  status = c("stub", "stub", "stub")
))
```

Table 1: Example QC summary (replace with your panel)

check	status
Rows	stub
Counties	stub
Date range	stub

7 Ethics and Responsible Use

- **Privacy:**
- **Fairness / equity:**
- **Limitations of setting:**
- **Non-causal framing:** *(if observational)*

8 Methods and Evaluation

Design:

Primary analysis:

Evaluation metrics:

Train / validation / holdout:

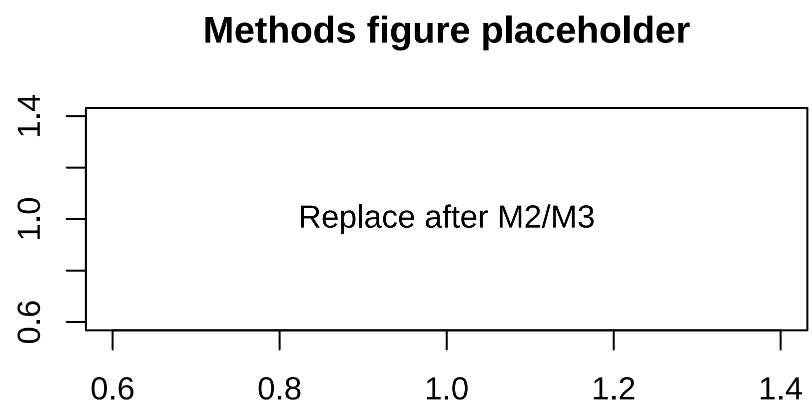

Figure 1: Replace with a methods diagram or exploratory plot

9 Results and Visualizations

(Stub: key findings with Figure 1 and tables.)

10 Discussion, Limitations, and Future Work

Interpretation:

Limitations: *(tie to actual data and design)*

Future work:

11 Conclusions and Recommendations

(Short, stakeholder-facing takeaways.)

12 Reproducibility and References

Reproduce main results:

```
cd your-repo
quarto render deliverables/M4-writeup-draft/writeup.qmd
```

Dependencies: `requirements.txt` or `renv.lock` in repo root.

Data: URLs and approval status documented in `docs/data-dictionary.md`.

13 References

Liu, J. C., L. J. Mickley, M. P. Sulprizio, et al. 2016. "Wildfire-Specific Fine Particulate Matter and Risk of Hospital Admissions in Urban and Rural Counties." *Epidemiology* 27 (3): 350–57.

Reid, C. E., M. Brauer, F. H. Johnston, et al. 2016. "Critical Review of Health Impacts of Wildfire Smoke Exposure." *Environmental Health Perspectives* 124 (8): 1334–43.